

Selection and Stability Issues Associated with a Navy Shipboard DC Zonal Electric Distribution System

John G. Ciezki, *Member, IEEE*, and Robert W. Ashton, *Member, IEEE*

Abstract—The U.S. Navy is currently investigating the implementation of a DC Zonal Electric Distribution System (DC ZEDS) for the next generation of surface combatant. In replacing the current ac radial distribution system, significant gains can be realized in terms of survivability, weight, manning and cost. DC ZEDS is predicated on having starboard and port dc busses feeding electrical zones delineated by watertight bulkhead compartments. The main bus dc voltage is stepped down within the zone and then converted to three-phase ac and lower voltage dc by additional power converters. Due to the large interconnection of tightly-regulated power converters in a stiffly-connected system, negative input impedance effects create the possibility of unwanted resonances. In this paper, the authors describe the rationale for the dc zonal system, characterize the stability issues, and discuss fault detection and load shedding problems.

Index Terms—dc power systems, marine vehicle propulsion, power distribution, stability.

I. SHIPBOARD DISTRIBUTION REQUIREMENTS

IN AN era of shrinking Department of Defense budget dollars, the U.S. Navy is actively engaged in developing cost-effective platforms that employ state-of-the-art and commercial-off-the-shelf technology to improve producibility, enhance operational flexibility, maximize survivability, and decrease manning requirements and overall cost [1]. With regards to shipboard electric power distribution, one obvious goal is to minimize the size and weight of the principle components in order to either shrink the size and signature of the platform or to augment the mission capability by inclusion of additional weapons systems or fuel.

A key design element of the power system is to provide alternate power paths to critical ship's service electrical loads such as combat systems, propulsion and fire-fighting resources. The system must be capable of seamlessly transferring power during fault conditions since a small glitch in power to a weapons system may have catastrophic results. The desirable quality of redundancy must be balanced against the requirements of low cost and compactness. The added flexibility and operating performance enabled by high-technology equipment must be balanced against the need to reduce manning onboard ship and maintain ruggedness in the presence of a hostile environment including shock, vibration and saltwater air. Finally, the electric distribution network should be adaptable to facilitate the integration of high-energy pulse-power weapons that are currently under development. In the following sections,

competing distribution architectures for a surface combatant vessel are discussed.

II. ZONAL VERSUS RADIAL DISTRIBUTION

Currently, surface combatants employ radial electric power distribution in which multiple generators (typically 3 or 4) are connected to a number of switchboard panels and the 450V, 60Hz three-phase ac is then distributed throughout the ship to load centers. The ship service generators are operated either in a split plant or a parallel configuration and are driven by either steam turbines, gas turbines or diesel engines. A representative gas turbine in current use is the 2500kW Allison 501. A U.S. Navy surface combatant has a ship service electrical load on the order of 3.3MW (normal cruising) to approximately 4MW (full combat cruising), though future equipment additions may extend that to the 5MW level. Each load is classified as being nonessential, semi-essential or essential. Therefore, if there is a loss of generation capacity, then a load shedding algorithm is initiated and circuit breakers are opened, disconnecting lower priority loads. Intelligent power management can help ensure that additional generators are not unnecessarily brought on-line to supply momentary increases in demand within the present capacity. Essential electrical loads onboard ship include emergency lighting and communications, fire pumps, search radar, compartment ventilation motors, prime mover auxiliary pumps and blowers, steering gear auxiliaries, and weapons systems. The generation capacity is deemed adequate if it can satisfy the operational requirements of the ship while at anchor, at pier side, cruising, performing its intended function and during emergency conditions.

Scores of three-phase step-down power transformers are used to establish 115V at the ship service convenience outlets and other voltages as needed. Both the transformer primary and secondary windings are connected in delta resulting in no reliable current path from the power lines to the ship's hull. The rationale for using a floating delta system revolves around survivability. If battle damage results in a single line being bolted to the ship's hull, large fault currents are prohibited from flowing since there is no complete low impedance path. As a result, protective devices in that section of the network will not be tripped and vital loads will not lose power during critical times. In addition, if a casualty results in the open-circuiting of one phase of a transmission line or in losing one leg of a transformer, balanced power can still be supplied, at a reduced capacity, by the open-delta connection.

The radial distribution architecture is contrasted with a zonal approach in Fig. 1. Note that the zonal approach employs a starboard and a port main bus and sections the ship into a number of

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The authors are with the Naval Postgraduate School, Monterey, CA 93943 USA.

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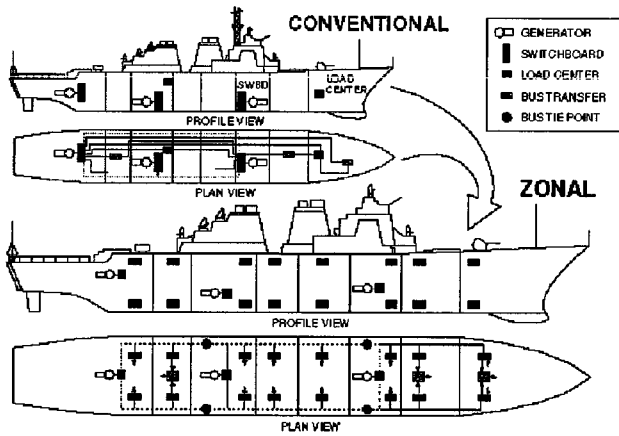


Fig. 1. Conventional and Zonal Distribution Architectures [2].

electrical zones that are delineated by physical watertight bulkhead compartments [2]. One main bus would be positioned well above the waterline while the other would be located below the waterline. This maximizes the distance between the busses and, in turn, maximizes the survivability of one of the busses during a casualty. An alternative architecture would place both busses along the ship centerline, displaced by several decks. This distribution architecture allows for maximum protection from battle damage near the skin of the ship, but would only be feasible for a larger-class ship.

Each bus connects into a zone via a load center and vital apparatus, such as combat systems, are connected to both load centers via an Automatic Bus Transfer (ABT) auctioneering process. The advantage of the zonal architecture is that now only the main busses transition watertight bulkheads and a myriad of switchboard feeder cables are eliminated [2]. This substantial reduction in cable cost and weight is accompanied by an additional benefit. Presently, cabling is installed onboard ships by manually pulling it through the various compartments. If cabling can be reduced, then this process is markedly simplified. In addition, a simplified distribution approach facilitates fabricating the ship in zones where the electrical cabling is installed as the zone is built. Such a producibility benefit offers considerable cost savings and allows for equipment installation and testing prior to zone assembly. A recent study performed on a USS Arleigh Burke class (DDG-51) guided missile destroyer platform indicated substantial weight and cost savings as documented in Tables I and II. The current architecture contains on the order of 200 switchboard feeder circuits and 400-circular-mil cables with some cables over 400 feet long. Note that most of the weight savings is achieved by removing 19,826 feet of power cable. A zonal architecture also reduces material and labor costs by \$1.373M per ship [2].

III. DC VERSUS AC ZONAL DISTRIBUTION

As mentioned above, ac zonal distribution is envisioned as consisting of bus/generator synchronizing and paralleling controls, two 60Hz primary busses, delta-delta connected transformers, load centers, and ac switchgear. An alternative architecture employs dc distribution as illustrated in Fig. 2

TABLE I
ZONAL VERSUS CONVENTIONAL ARCHITECTURE WEIGHT COMPARISON [2]

Apparatus	Removal (LT)	Install (LT)	Net Change (LT)
Foundations	3.3	4.3	+1.0
Power Cables	116.7	79.8	-36.9
Switchgear	20.8	20.0	-0.8
Total	140.8	104.1	-36.7

TABLE II
ZONAL VERSUS CONVENTIONAL ARCHITECTURE COST COMPARISON [2]

Apparatus	Conventional Material+Labor Total	Zonal Material+Labor Total
Foundations	34k	44k
Power Cables	4,151k	2,839k
SWBD+LC*	1,807k	1,736k
Total	5,992k	4,619k

*Switchboard and Load Centers

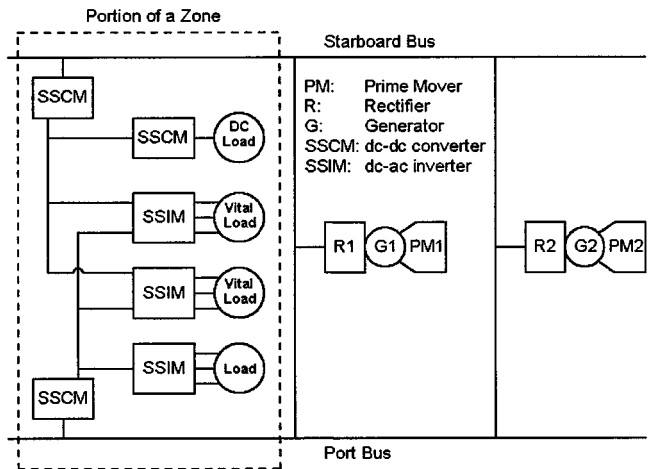


Fig. 2. DC Zonal Architecture.

(DC ZEDS). Note that in this case, the generator voltages are rectified and the resulting dc, on the order of 1200 to 2000V, is distributed along the port and starboard busses. Each bus is connected to an electrical zone through a power converter that serves to buffer the main bus and the intra-zone electrical loads. In addition, the converter adjusts the main bus voltage downward to one commensurate with dc-ac inverter requirements (on the order of 850 to 950V) and other dc loads. The inverters would then provide the single and three-phase power requirements of the zone. Paralleled converters and inverters would be used to establish the required power levels and guarantee the needed redundancy.

The input buffering converter is a step-down dc-dc buck converter and is termed the Ship Service Converter Module (SSCM) while the dc-ac pulse-width-modulation (pwm) inverter unit is termed the Ship Service Inverter Module (SSIM). Resonant converter topologies, such as the Auxiliary Resonant

Commutated Pole (ARCP) converter, are being considered to minimize losses and facilitate higher switching-frequencies, which in turn reduce airborne audible noise, maximize control bandwidth and minimize filtering requirements. The standardized converter modules are anticipated to extend across multiple classes of ships, reducing ship acquisition costs and enhancing the ability to reassign sailors to different classes of ships without retraining.

There are several advantages inherent to dc distribution. First, the SSCM's and SSIM's are multifunctional, performing the power conversion, monitoring and limiting the current through the semiconductor devices, and suitably protecting the system during fault conditions. Thus, DC ZEDS facilitates isolating faults to an electrical zone. Since the inputs to each are dc currents, the current sensors and algorithms required to detect fault conditions are both simpler and faster. As a result, the time lag associated with detecting ac faults and initiating the Automatic Bus Transfer (ABT) is virtually eliminated and the integrity of the power to vital loads is enhanced. With improved continuity of power and automated power management, fewer uninterruptible power supplies may be required and the design of combat systems simplified.

A second advantage of DC ZEDS is that variable speed motor control is readily available to the many pumps and blowers onboard to ensure operation of these devices at the highest efficiencies. In addition, the substantial in-rush currents experienced when starting large motors may be limited, aiding in the maintenance of a stable bus voltage.

In certain instances, DC ZEDS will also offer an advantage in terms of the number of power conversion stages. For instance, various combat systems require 400Hz together with a number of dc voltages. In the ac distribution system, this would require that the distributed three-phase ac must be rectified, converted to 400Hz with an inverter, shifted to an appropriate voltage level with a transformer, then once again rectified to provide the required dc power. In DC ZEDS, there is no need to have an intermediate 60Hz step. Also, it is clear that most of the distribution transformers and ac switchgear may be eliminated in DC ZEDS. This may offer a considerable benefit in terms of both weight and size. The gain in space allows the ship designer to re-orient compartments to include additional weapons systems, cargo and/or fuel to enhance the mission capability.

Another advantage of DC ZEDS is that the generator frequency is decoupled from the distribution requirements. Therefore, the generator/rectifier combinations may be optimized for size and cost. In addition, the prime movers may also be designed so that they always run at the most efficient speed. This, in turn, has the subsidiary benefits of optimizing fuel consumption and reducing emissions.

One final advantage inherent to DC ZEDS is that it employs state-of-the-art semiconductor switch technology which is engendered by the Office of Naval Research Power Electronic Building Block (PEBB) program. As advances in the field are realized and new converter components are developed, DC ZEDS, versus AC ZEDS, significantly facilitates ready inclusion of new devices.

The disadvantages of dc distribution are primarily associated with the main bus voltage. Currently, SSCM and SSIM tech-

nology are based on insulated-gate-bipolar-transistor (IGBT) and MOS-controlled-thyristor (MCT) power semiconductor devices that have voltage ratings on the order of 1500 to 2000V. This limits the admissible bus voltage. Obviously, higher voltages are preferred to reduce the size and weight of the transmission line cabling, though this introduces grounding requirements and isolation issues with the SSCM's. In addition, pulse-power Electro-Thermal Chemical (ETC) guns require dc voltage levels on the order of 15 000V [3]-[4]. If 15 000V dc was the distribution level, then these weapons could potentially be supplied directly. One final complication of DC ZEDS involves operation of the system at dock when shore power, typically 450V three-phase, is available. A regulated power converter interface would be required to establish and maintain the required dc bus voltage. Regardless, the substantial gains in survivability, size and weight, efficiency and enhanced performance, and improved flexibility motivate the selection of DC ZEDS over an ac counterpart.

IV. INTEGRATED POWER SYSTEM

Currently, surface combatants employ either a fixed-pitch or a control lable-reversible-pitch (CRP) propeller. Speed control is achieved by governed manipulation of the gas turbine throttle input and, if available, adjustment of the CRP propeller. If a CRP is not used, the mechanical drive propels the ship in reverse by employing reversing gears or reversing turbines. Additional maneuvering capability at port or when the propulsion system suffers a casualty is provided by two Auxiliary Propulsion Units (APU's) which are lowered down into the water from the bottom of the ship. Each unit is driven by two electric motors supplied from the ship service distribution system.

Integrated Power System (IPS) is the term applied to a ship architecture where both ship service loads and ship propulsion are supplied from a common electrical source. The concept of marine electric propulsion dates back to the early 1900's and early examples include the U.S. battleships *New Mexico* and the carrier *Saratoga* [5]. More recently, various cruise line ships and icebreakers have gone to modern electric drives for many of the reasons outlined below [6]-[7].

From a naval architecture perspective, electric drive offers a number of advantages over the current mechanical drive [8]:

- the number of prime movers is reduced;
- the prime mover is not required to be directly in line with the propeller shaft;
- through suitable electrical cross-connect, any prime mover can power any propulsion motor, minimizing killability;
- the prime mover speed is decoupled from the propeller speed;
- infinitely variable speed control enhances maneuverability and simplifies propeller reversals;
- and gear train noise is eliminated, reducing ship signature.

Subsidiary benefits of the above include:

- fewer prime movers imply fewer maintenance requirements;
- supply and exhaust ducting can be minimized by placing prime movers on higher decks;

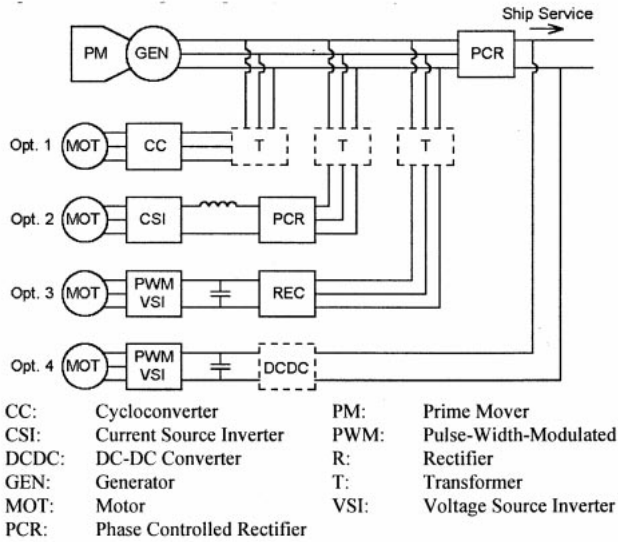


Fig. 3. Electric Drive Topology Options

- a shorter shaft line reduces the complexity of equipment alignment across ship construction zones;
- reversing gears/turbines are eliminated and a CRP propeller becomes optional, which reduces both cost and cavitation concerns;
- and prime mover operation is optimized, maximizing fuel efficiency and minimizing emissions.

The Naval Sea Systems Command Advanced Surface Machinery program estimates in one study that IPS can reduce ship acquisition cost by \$12.8M, reduce annual fuel costs by 24% in comparison to a similar mechanical drive, reduce full-load displacement by 14%, and reduce engineering department manning by 13% [8].

A surface combatant electric drive requires a high-power (20MW), low-speed (0 to 220rpm) machine/converter unit that is as compact as possible. Since the market for such a unique drive is small, the U.S. Navy must specifically contract its design, which increases research costs and lengthens development time-lines. An additional obstacle to electric drive is breaking down the institutional hesitancy to incorporating advanced technologies and demonstrating the required system robustness. The feasibility of electric drive hinges on the development of high power density electric machinery and power converters [9],[13]. Ongoing research into permanent magnet machinery [10]-[11] and superconducting homopolar machinery [12] is seeking to address the density issue, though machinery at the 20MW level is not currently available.

Several IPS architectures are possible [18] and Fig. 3 illustrates four ac motor propulsion system options. As shown, the first three options require a multi-phase ac cable feed from the synchronous generator to the propulsion converters. The fourth option illustrates a feed from the dc distribution bus to the propulsion converter. Components shown in dotted boxes are required only if motor/generator/converter rated voltages are not suitably matched. Cycloconverter and current source inverter propulsion drives are currently in use onboard

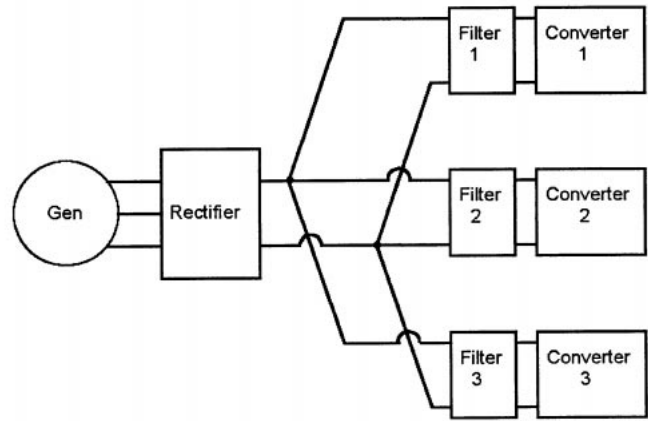


Fig. 4. Interconnection of High-Bandwidth, Nonlinear Power Converters.

U.S. Coast Guard icebreakers and passenger cruise line ships [14]-[16]. Pulse-width-modulated voltage source inverter drives based on current IGBT or MCT technology are limited in power rating to about 2MW. As a consequence, these converters would either need to be stacked or new topologies, such as multi-level converters, employed in order to satisfy the 20MW propulsion motor demands. The propulsion motor could be a multi-phase induction machine or a field-wound synchronous machine though, as stated previously, intense work in ongoing at developing a permanent magnet machine option at the required power level.

Current technology is available for the design and development of a surface combatant employing an electric drive. Future technological advancements in electric machinery and power converters are expected to further diminish the power density concerns and provide additional engineering design flexibility.

V. DC DISTRIBUTION STABILITY ISSUES

The DC Zonal Electric Distribution System described in Section III introduces several stability issues related to the interconnection of a number of high-bandwidth, nonlinear power converters as shown in Fig. 4 [17]. Since it is possible that the system may be reconfigured due to casualty or an intentional system expansion, understanding the stability questions and guaranteeing the integrity of the system operation is vital. The stability of ac systems, whether stiffly connected or following an infinite-bus assumption, is well understood. Less is known about networks of power converters in a stiffly-connected platform. The primary issue may be described using a simple example. In terms of the relatively slow dynamics of the source, the networked high-bandwidth power converters may be modeled by equivalent input filters and a constant power load as illustrated in Fig. 5. Upon establishing the modeling equations for this simple nonlinear network, it is found that a constraint is placed on the source resistance in order for there to be a steady-state operating point (stability). For the case illustrated in Fig. 5,

$$R_S < \frac{V_S^2}{4(P_1 + P_2 + P_3)}.$$

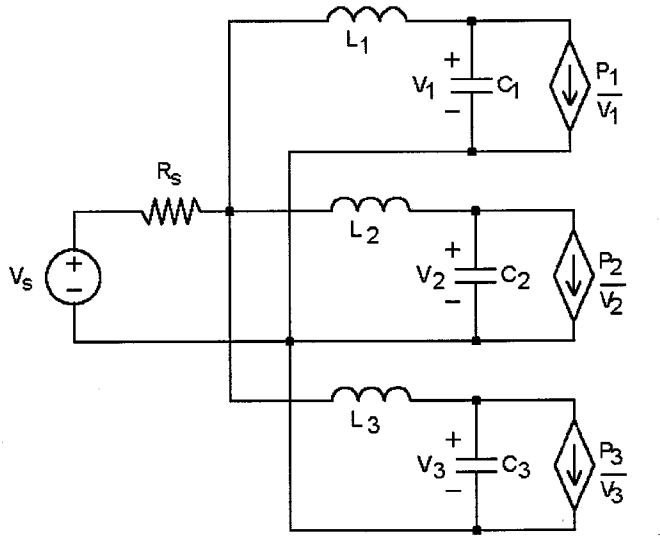


Fig. 5. Model of Networked High-Bandwidth Converters.

Now there will also be a lower bounding constraint based on avoiding a sustained oscillation or limit cycle in the network. The determination of this requirement is more involved and will change as the network configuration and load changes [17]. Certainly this is an addressable problem, though it imposes important implications on the design of the individual power converter components. Clearly, a criterion on input impedance must be assigned for each device that guarantees that if that device is inserted into the network, in a myriad of possible configurations, the system will remain stable. The issue of dc system stability is an ongoing research area in DC ZEDS in terms of system characterization, analysis and control.

VI. CONCLUSIONS

The authors have discussed current and potential distribution and propulsion options for U.S. surface combatant vessels. The advantages of zonal and dc zonal distribution were delineated and the dc stability issues introduced. The advantages of an integrated power system employing an electric drive were contrasted with the current technical hurdles. With improvements in semiconductor devices, including new technologies such as silicon-carbide, new converter architectures, such as multi-level converters, and electric machinery design advances, such as in permanent magnet and homopolar machines, a suitable integrated power system candidate for a naval surface combatant will emerge.

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John G. Ciezki received his B.S.E.E., M.S.E.E., and Ph.D. from Purdue University in 1988, 1990, and 1993, respectively. While working on his degrees, he has done extensive research for P. C. Krause and Associates in electric drive modeling and simulation. Following post-doctorate work at Purdue, he joined the staff at the Naval Postgraduate School in 1994 as an Assistant Professor in the area of Power Systems.

Robert W. Ashton graduated from Virginia Polytechnic Institute with a B.S.E.E. in 1982. He worked for Unisys Corporation as a Component Engineer until returning to school in 1986. He received his M.S.E.E. in 1989 from Worcester Polytechnic Institute. He earned a Ph.D. in the area of power electronics while a fellow with Northeast Utilities in Connecticut. He currently holds the position of Associate Professor in the Department of Electrical and Computer Engineering at the Naval Postgraduate School in Monterey, CA.